NUMBER & ALGEBRA

Chapter 1

Exponents & Logarithms

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SYLLABUS 1.1 1.5 1.7

Japan lies in one of the world's most active seismic zones. Suppose a seismologist asks you to plot two recordings on one axis. The first has an amplitude (the height of the wave) of 10^4. The second has an amplitude of 10^9, which is $10^5 = 100{,}000$ times the first.

The task sounds simple, but an ordinary scale, with evenly spaced marks, cannot show both. If the marks are fine enough for the first recording, the second falls far beyond the page. If the scale is long enough for the second, the first sits so close to zero that you cannot see it.

Now suppose equal spaces along the axis meant multiplication by 10 rather than addition of 1. The scale would read $10^0, 10^1, 10^2$, and so on. The two recordings have exponents 4 and 9, so they would sit five equal steps apart, and both would fit on one page. That exponent has a name. It is the **logarithm** of the amplitude, and it is the idea this chapter is built on.

This is how earthquake magnitude scales work. One step up in magnitude means ten times the recorded amplitude. Writing a quantity as a power of ten turns multiplication into addition, and that is what fits an enormous range into a small space. The same idea gives us pH, decibels, and the scale for the brightness of stars. It is useful because the hard operation becomes the easy one: quantities that would have to be divided can instead be subtracted, and a range too wide to plot fits on an axis you can read.

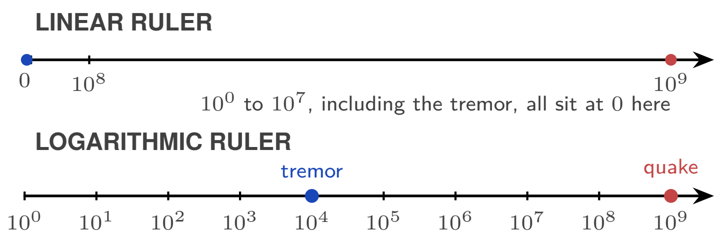

Figure 1.1 A logarithmic scale separates values that collapse near zero on a linear scale.

By the end of this chapter you will be able to use the laws of exponents, write quantities in standard form, and move between exponential and logarithmic form. You will also solve equations in which the unknown sits in an exponent. Each of these does one job: it puts a quantity into the form that makes a comparison easy. That is why a scale built on powers of ten can hold the mass of an electron and the mass of a star on one axis.

1.1 What an Exponent Counts

An exponent, also called a *power* or an *index*, is shorthand for repeated multiplication. In a^n, a is the **base** and n is the **exponent**. When n is a positive integer,

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}}.$$

For example, $a^3 = a \cdot a \cdot a$.

When you multiply powers with the same base, it is tempting to multiply the exponents. Writing out the factors shows what actually happens:

$$a^3 \cdot a^2 = (a \cdot a \cdot a)(a \cdot a) = a^5.$$

There are five factors of a, not six. Since the exponent counts how many times the base appears as a factor, the exponents are **added**:

$$a^m \cdot a^n = a^{m+n}.$$

Division cancels common factors, so the exponents are **subtracted**:

$$\frac{a^5}{a^2} = \frac{a \cdot a \cdot a \cdot a \cdot a}{a \cdot a} = a^3 = a^{5-2}.$$

Raising a power to another power repeats the whole power, so the exponents are **multiplied**:

$$(a^2)^3 = a^2 \cdot a^2 \cdot a^2 = a^6 = a^{2\times 3}.$$

A power outside a bracket applies to every factor inside:

$$(ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

The same laws extend naturally to zero and negative exponents. For $a \neq 0$, the subtraction rule gives $a^3/a^3 = a^{3-3} = a^0$; since any nonzero number divided by itself is 1, this forces $a^0 = 1$. Continuing the same pattern gives $a^{-n} = 1/a^n$. These values are not imposed on us; they are the only ones that keep the laws consistent. The main laws are collected below.

DEF For $a, b \neq 0$ and $m, n \in \mathbb{Z}$:

$a^m \cdot a^n = a^{m+n}$	product of powers: add the exponents
$\dfrac{a^m}{a^n} = a^{m-n}$	quotient of powers: subtract the exponents
$(a^m)^n = a^{mn}$	power of a power: multiply the exponents
$(ab)^n = a^n b^n$	power of a product: raise each factor
$\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$	power of a quotient: raise top and bottom
$a^0 = 1$	zero exponent: gives 1
$a^{-n} = \dfrac{1}{a^n}$	negative exponent: take the reciprocal

CAUTION *Do not confuse $(a^m)^n$ with a^{m^n}. In $(a^m)^n$, the power of a power, you multiply the exponents. In a^{m^n} you start at the top: first work out m^n, then raise a to that result. The two are usually not equal: $(2^3)^2 = 2^6 = 64$, but $2^{3^2} = 2^9 = 512$.*

CAUTION *A negative exponent means reciprocal, not negative value. $2^{-3} = \frac{1}{8}$, not -8.*

Worked examples

How do these rules combine when a problem has several operations at once?

EXAMPLE 1.1

Simplify $\dfrac{6c^9}{2c^4}$.

Split the coefficient and the variable and handle them separately.

$$\frac{6c^9}{2c^4} = \frac{6}{2} \cdot c^{9-4} = 3c^5.$$

Apply one rule at a time.

EXAMPLE 1.2

$(2x^2y^{-4})^5$ contains several operations: a bracket raised to a power and a negative exponent. Raise each part in turn: the 2 to the 5th, x^2 to the 5th, and y^{-4} to the 5th.

$$(2x^2y^{-4})^5 = 32x^{10}y^{-20}.$$

> The most common error here is forgetting to raise the coefficient.

CAUTION *The bracket in* $(2x^2)^3$ *applies to everything inside it, including the* 2. $(2x^2)^3 = 8x^6$, *not* $2x^6$.

Beyond counting factors

The idea of a^n as repeated multiplication only makes sense when n is a positive integer. For example, what could 2^{-3} mean? You cannot multiply 2 by itself -3 times. And what is $3^{1/2}$, or 4^0? You cannot take half a factor, or no factors at all. Yet each of these still has exactly one sensible value. We choose that value so the exponent laws keep working:

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}, \qquad 3^{1/2} = \sqrt{3}, \qquad 4^0 = 1.$$

The next sections show where these values come from.

Surds and fractional exponents

What is $9^{1/2}$? Since $\frac{1}{2}$ is not an integer, repeated multiplication cannot answer this directly. But the power-of-a-power law gives

$$\left(9^{1/2}\right)^2 = 9^{(1/2)\times 2} = 9,$$

so $9^{1/2}$ must be the non-negative number whose square is 9:

$$9^{1/2} = \sqrt{9} = 3.$$

Notice the value is 3, not ± 3: the symbol $\sqrt{9}$ means the **principal** (non-negative) square root.

DEF For $a > 0$, $n \in \mathbb{Z}^+$ and $m \in \mathbb{Z}$:

$a^{1/n} = \sqrt[n]{a}$	unit fraction: the denominator is the root
$a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$	in general: root then power, in either order

A useful way to remember this:

$$\boxed{\text{denominator } n \longrightarrow \text{root}, \qquad \text{numerator } m \longrightarrow \text{power}}$$

These are not new laws; they are the meanings chosen so that the familiar exponent laws keep holding.

Using fractional exponents

For $a^{m/n}$, it is usually easier to take the root first, then apply the power:

$$64^{2/3} = \left(\sqrt[3]{64}\right)^2 = 4^2 = 16.$$

Either order gives the same result, $64^{2/3} = \sqrt[3]{64^2} = 16$, but taking the root first keeps the numbers small.

Working with surds

A **surd** is an irrational root left in exact form, such as $\sqrt{2}$ or $\sqrt[3]{5}$, rather than rounded to a decimal. On a non-calculator paper, surds should be kept exact and simplified using the root laws

$$\sqrt{ab} = \sqrt{a}\,\sqrt{b}, \qquad \sqrt{\tfrac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}},$$

valid wherever the quantities are real. Two techniques cover most questions:

1. **Simplify** a surd by extracting its largest perfect-square factor.
2. **Combine** only like surds, the terms that have the same value under the root.

Rationalising the denominator

An exact answer is usually written without a surd in the denominator; removing it is called **rationalising the denominator**.

- For a single surd, multiply top and bottom by that surd.
- For a sum or difference, multiply by its **conjugate**, formed by changing the sign between the terms.

EXAMPLE 1.3

Rationalise $\dfrac{1}{\sqrt{3}}$ and $\dfrac{6}{2+\sqrt{3}}$.

For the first, multiply by $\dfrac{\sqrt{3}}{\sqrt{3}}$:

$$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}.$$

For the second, multiply by the conjugate $2-\sqrt{3}$:

$$\frac{6}{2+\sqrt{3}} = \frac{6(2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})} = \frac{6(2-\sqrt{3})}{4-3} = 12 - 6\sqrt{3}.$$

The conjugate works because the product is a difference of two squares, $(2+\sqrt{3})(2-\sqrt{3}) = 2^2 - (\sqrt{3})^2 = 1$, so the surd leaves the denominator.

TIP *In the real numbers, $(-4)^{1/2}$ is undefined: no real number squares to -4. In general, the radicand, the number inside the root, must be non-negative for an even root to have a real value. Square roots of negative numbers are complex numbers, which are introduced in Chapter 18.*

Exponents as continuous growth

There is another way to read a non-integer exponent. Suppose 2^t means growth by a factor of 2 over a time t. Then 2^3 is three complete doublings, while

$$2^{1/2} = \sqrt{2}$$

is the growth factor halfway through one doubling period. Because the growth is exponential, "halfway" refers to the time, not to half the final increase. This reading extends to every real value of t, whether negative, fractional, or irrational:

$$2^{-3}, \qquad 2^0, \qquad 2^{1/2}, \qquad 2^{\sqrt{2}}.$$

Read as growth over time t, each has a plain meaning. $2^0 = 1$ is no growth at all: at time zero the population is unchanged, one hundred per cent of where it started. $2^{-3} = \frac{1}{8}$ looks backwards in time, to three periods before now (three seconds ago, if one doubling takes a second), when the population was an eighth of its present size. $2^{1/2} = \sqrt{2}$ is the factor after half a period, and $2^{\sqrt{2}}$ the factor after an irrational stretch of time. For any positive base a, the expression a^x has a meaning for every real exponent x, which is where the exponential function of Chapter 6 begins.

The number *e*

In the previous section, 2^t described growth that doubles every period. But real growth is usually smooth. A population does not wait until the end of the hour to add all its new members; it grows a little at every moment. What number describes growth that happens continuously, at every instant?

Take a quantity that would grow by 100% in one hour. If all of that growth arrives at the very end, the quantity simply doubles. But suppose the same 100% is spread over two half-hours. After the first half-hour it is $1 \times 1.5 = 1.5$ times its start, and that larger amount then grows by the next 50%, reaching $1.5 \times 1.5 = 2.25$ times. Spreading the growth into more steps lets each new part start growing sooner, so the hour's total rises a little each time. In n equal steps the quantity multiplies by $\left(1 + \frac{1}{n}\right)^n$:

steps in the hour, n	**size after one hour**
1 (once)	2.000
2 (half-hourly)	2.250
60 (every minute)	2.6960
3600 (every second)	2.7179
$n \to \infty$ (continuous)	$2.71828\ldots = e$

The totals do not grow without bound. They approach a single limit, $2.71828\ldots$, and this number is called e. Like π, it cannot be written as a finite decimal, and it appears throughout mathematics. Here it is the result of 100% growth acting continuously, at every instant instead of in fixed steps. A continuous rate r acting for time t multiplies the starting amount by e^{rt}. This is the model behind population growth, radioactive decay, and cooling. Chapter 14 gives a deeper reason for e: the function e^x grows at a rate equal to its own value.

EXT Formally, $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$. Each n is a number of compounding steps per year, so letting n grow without bound is what "continuously" means. Limits are defined properly in Ch. 14.

YOUR TURN 1.1 What an Exponent Counts

1. Simplify each of the following.

(a) $\dfrac{6c^9}{2c^4}$ [1] (b) $(3x^2y^{-3})^4$ [2]

(c) $\dfrac{12a^5b^3}{4a^2b^{-1}}$ [2] (d) $\dfrac{(2x^3)^2\cdot x^{-4}}{4x^2}$ [2]

2. Simplify by dividing each term of the numerator separately.

(a) $\dfrac{4x^3-8x^2}{2x}$ [2] (b) $\dfrac{16a^5b^2-12ab^4}{4ab^2}$ [2]

3. Show that $\dfrac{(a^2b^{-1})^3}{a^{-3}b^2} = \dfrac{a^9}{b^5}$. [3]

4. Evaluate each of the following.

(a) $27^{2/3}$ [1] (b) $81^{-3/4}$ [2]

(c) $\left(\dfrac{8}{27}\right)^{-2/3}$ [2] (d) $32^{-2/5}$ [2]

5. Write each of the following in the form x^n.

(a) $\sqrt[4]{x^3}$ [1] (b) $\dfrac{1}{\sqrt[3]{x^2}}$ [1]

(c) $\dfrac{\sqrt{x^5}}{x^2}$ [2] (d) $\dfrac{x^2\cdot\sqrt{x}}{x^{3/2}}$ [2]

6. Solve $x^{3/2} = 64$. [2]

7. Simplify each of the following.

(a) $\sqrt{50}-\sqrt{8}$ [1] (b) $\sqrt{12}+\sqrt{27}$ [2]

8. Rationalise the denominator of $\dfrac{4}{3-\sqrt{5}}$, giving your answer in the form $a+b\sqrt{5}$. [3]

1.2 Standard Form

The mass of a proton, written out in full, is

$$0.00000000000000000000000000167 \text{ kg},$$

that is, 26 zeros after the decimal point before the first significant figure. The mass of the Sun runs to the opposite extreme,

$$1{,}990{,}000{,}000{,}000{,}000{,}000{,}000{,}000{,}000{,}000 \text{ kg},$$

a 1 followed by 30 digits. Both numbers are quite difficult to write, let alone compare or calculate with.

There is a shorter way to write both. Instead of all the digits, record *the power of 10* that places the decimal point where it belongs. Mathematicians call the result **standard form** (also known as scientific notation).

DEF A number is in **standard form** when written as $a \times 10^k$, where $1 \leq a < 10$ and $k \in \mathbb{Z}$.

The proton becomes 1.67×10^{-27} kg. The Sun becomes 1.99×10^{30} kg. The exponent k tells you immediately whether you are dealing with something microscopic or cosmic.

Calculating with standard form

Multiplication and division follow straight from the exponent laws. For a product, multiply the coefficients and add the powers of 10; for a quotient, divide the coefficients and subtract the powers:

$$(a \times 10^m)(b \times 10^n) = (a \times b) \times 10^{m+n}.$$

One check matters afterwards. If the new coefficient falls outside the range $1 \leq a < 10$, shift the decimal point and adjust the power. For example, $(5 \times 10^3)(4 \times 10^2) = 20 \times 10^5 = 2 \times 10^6$: the coefficient 20 becomes 2, and the exponent rises by one to account for the extra factor of 10.

Addition and subtraction need one more move first. Just as you cannot add metres and kilometres without converting, you cannot add two numbers in standard form until they share the same power of 10. Rewrite the one with the smaller exponent so the powers match, then add the coefficients.

EXAMPLE 1.4

Calculate $(3.5 \times 10^4) \times (2 \times 10^{-7})$.

Group and multiply:

$$(3.5 \times 2) \times 10^{4+(-7)} = 7 \times 10^{-3}.$$

EXAMPLE 1.5

Calculate $(4.8 \times 10^5) + (3.2 \times 10^4)$.

Before adding, you need the same power of 10, just as you cannot add metres and kilometres without converting first.

$$4.8 \times 10^5 + 0.32 \times 10^5 = 5.12 \times 10^5.$$

CAUTION *In addition and subtraction, do **not** add the exponents. Only multiplication and division combine powers of 10.*

CAUTION *Calculator display notation is not acceptable in written answers. If your graphic display calculator (GDC) shows 5.2E30, you must write 5.2×10^{30}.*

YOUR TURN 1.2 Standard Form

9. Write 0.000047 in standard form. [1]

10. Write 3.2×10^5 as an ordinary number. [1]

11. Calculate $(2.4 \times 10^3) \times (3 \times 10^{-5})$, giving your answer in standard form. [2]

12. Calculate $\dfrac{6.6 \times 10^8}{2.2 \times 10^3}$, giving your answer in standard form. [2]

13. Calculate $(5.1 \times 10^4) + (3.9 \times 10^3)$, giving your answer in standard form. [2]

14. GDC The mass of an electron is 9.11×10^{-31} kg. The mass of a proton is 1.67×10^{-27} kg. Find how many times heavier a proton is than an electron. Give your answer in standard form to 3 significant figures. [2]

1.3 The Logarithm: an Exponent Reversed

Many familiar operations come in inverse pairs: addition with subtraction, multiplication with division. So what is the inverse of exponentiation? If $a^x = b$, and you know a and b, how do you find x? That is what a logarithm does. It is not a new idea: it is the exponential equation $a^x = b$ rearranged to make the exponent x the subject.

DEF · IB For $a > 0$, $a \neq 1$, and $b > 0$:

$$a^x = b \iff x = \log_a b$$

The **common logarithm** ($\log x$) uses base 10. The **natural logarithm** ($\ln x$) uses base $e \approx 2.718$, the continuous-growth constant met earlier in this chapter.

The word *logarithm* comes from the Greek *logos* (ratio) and *arithmos* (number).

The graph of $y = \log_a x$ is the mirror image of $y = a^x$ across the line $y = x$, with a vertical asymptote at $x = 0$. These graphs and their transformations are developed in Chapter 6 (Functions).

Because the exponential and logarithm are inverses, each undoes what the other does:

KEY For base 10 and base e:

$\log(10^x) = x$	base 10: true for every real x
$10^{\log x} = x$	the same pair reversed, but only for $x > 0$, since $\log x$ is undefined otherwise
$\ln(e^x) = x$	base e: true for every real x
$e^{\ln x} = x$	reversed, and again only for $x > 0$

Two special values follow at once from the definition, and are worth knowing on sight: since $a^1 = a$ and $a^0 = 1$,

KEY For any base $a > 0$, $a \neq 1$:

$\log_a a = 1$	since $a^1 = a$
$\log_a 1 = 0$	since $a^0 = 1$

Working with logarithms

Evaluating a logarithm by hand is always the same move in reverse. Since $\log_a b$ is defined as the exponent that turns a into b, every logarithm question is an exponent question asked backwards. Read $\log_a b$ as **"a to what power gives b?"** and in many cases the answer is immediate.

EXAMPLE 1.6

Find $\log_3 81$.

Ask the inverse question: 3 to what power gives 81? You know $3^4 = 81$, so $\log_3 81 = 4$. The logarithm is just the exponent.

EXAMPLE 1.7

Solve $\log_{10}(y+1) = 3$.

The equation says: the exponent you need to get $(y+1)$ from base 10 is 3. So $y+1 = 10^3 = 1000$, giving $y = 999$.

EXAMPLE 1.8

Without a calculator, find $e^{3\ln 2}$.

Think of $e^{\ln 2}$ first: the exponential and natural log cancel, giving 2. So $e^{3\ln 2} = (e^{\ln 2})^3 = 2^3 = 8$.

Laws of logarithms

Where the laws come from

Why do the logarithm laws work? A logarithm is an exponent. When you multiply two powers with the same base, you add the exponents: $a^p \cdot a^q = a^{p+q}$. In terms of logarithms, this means that multiplication becomes addition.

The first three laws below are the three exponent laws restated for logarithms. A fourth, change of base, is a different kind of rule: it rewrites any logarithm as a ratio of two logarithms in a base of your choice.

KEY · IB For $a, b, x, y > 0$ with $a, b \neq 1$:

$\log_a(xy) = \log_a x + \log_a y$	product: a product becomes a sum
$\log_a\left(\dfrac{x}{y}\right) = \log_a x - \log_a y$	quotient: a quotient becomes a difference
$\log_a(x^m) = m\log_a x$	power: an exponent becomes a coefficient
$\log_a x = \dfrac{\log_b x}{\log_b a}$	change of base: rewrite in any base you can compute

All three proofs share one setup. Write $x = a^p$ and $y = a^q$, so that $p = \log_a x$ and $q = \log_a y$ by definition. Each law is then a one-line consequence of the matching exponent law:

- **Product.** $xy = a^p \cdot a^q = a^{p+q}$, so $\log_a(xy) = p + q = \log_a x + \log_a y$.
- **Quotient.** $\dfrac{x}{y} = a^{p-q}$, so $\log_a\left(\dfrac{x}{y}\right) = p - q = \log_a x - \log_a y$.

- **Power.** $x^m = (a^p)^m = a^{mp}$, so $\log_a(x^m) = mp = m\log_a x$.

The third law is the most useful in this course. When x appears in an exponent, the power law $\log_a(x^m) = m\log_a x$ moves the exponent into a coefficient. This often converts an exponential equation into a simpler algebraic equation.

CHANGE of base

What do you do when your GDC only has log (base 10) and ln (base e), but you need a logarithm in a different base, or you want to rewrite a base-10 logarithm in terms of base e? The change-of-base law above rewrites any logarithm as a ratio of two logarithms you can compute. In practice you take base e, which turns the law into $\log_a x = \dfrac{\ln x}{\ln a}$.

The proof is elegant: let $\log_a x = y$, so $x = a^y$. Take ln of both sides: $\ln x = y\ln a$. Solve for y.

> **EXAMPLE 1.9**
>
> Evaluate $\log_3 5$ on a GDC.
>
> Convert to natural logarithms:
>
> $$\log_3 5 = \frac{\ln 5}{\ln 3} \approx \frac{1.609}{1.099} \approx 1.46.$$

APPLYING the laws

These laws can be used in either direction: they split one logarithm into several, or gather several back into one. Both directions appear constantly, in simplifying expressions and in solving equations.

> **EXAMPLE 1.10**
>
> Express $\log_5\left(\dfrac{a^3}{b}\right)$ in terms of $p = \log_5 a$ and $q = \log_5 b$.
>
> Apply the quotient law, then the power law:
>
> $$\log_5\left(\frac{a^3}{b}\right) = \log_5 a^3 - \log_5 b = 3\log_5 a - \log_5 b = 3p - q.$$

> **EXAMPLE 1.11**
>
> Write $2\ln a + 6\ln b$ as a single logarithm.
>
> Use the power law in reverse first, then the product law:
>
> $$2\ln a + 6\ln b = \ln a^2 + \ln b^6 = \ln(a^2b^6).$$

> **EXAMPLE 1.12**
>
> Solve $\log_2 x + \log_2(x-2) = 3$.
>
> The two logs on the left can combine into one product:
>
> $$\log_2[x(x-2)] = 3 \implies x(x-2) = 2^3 = 8.$$

$$x^2 - 2x - 8 = 0 \implies (x-4)(x+2) = 0.$$

So $x = 4$ or $x = -2$. But $\log_2(-2)$ does not exist, so reject $x = -2$. The answer is $x = 4$. Always check domain when logarithms are involved. It is easy to generate solutions that look valid algebraically but are meaningless.

CAUTION $\log(x+y) \neq \log x + \log y$. *The product law is about* $\log(xy)$, *not a sum inside the log.*

Solving exponential equations

Same base first

Is there ever a way to solve an exponential equation without using a logarithm at all? Before anything else, look for a common base. If both sides can be written as powers of the same number, the exponents must be equal, provided the base is not 0 or 1: $a^p = a^q$ forces $p = q$ for $a > 0$, $a \neq 1$. This is the fastest route whenever it is available, and the intended one on non-calculator papers.

EXAMPLE 1.13

Solve $\left(\frac{1}{3}\right)^x = 9^{x+1}$.

Both sides are powers of 3: $\frac{1}{3} = 3^{-1}$ and $9 = 3^2$. Rewrite and equate exponents:

$$3^{-x} = 3^{2(x+1)} \implies -x = 2x + 2 \implies x = -\tfrac{2}{3}.$$

No logarithm was needed. Check for a shared base before anything else; look for common bases such as $2, 4, 8$ and $3, 9, 27$.

The general strategy

When no common base exists, take a logarithm of both sides, use $\log(x^m) = m \log x$ to bring the exponent down, then solve what is now a linear or quadratic equation. The choice of which logarithm to use (log or ln) does not matter: pick whichever gives cleaner arithmetic.

EXAMPLE 1.14

Solve $5^x = 30$.

Take ln of both sides:

$$x \ln 5 = \ln 30 \implies x = \frac{\ln 30}{\ln 5} \approx 2.11.$$

Or equivalently, $x = \log_5 30$.

EXAMPLE 1.15

Solve $7^{x-2} = 5^{x+3}$ exactly.

With different bases on each side, take ln of both sides and use the power law:

$$(x-2)\ln 7 = (x+3)\ln 5.$$

Collect x on one side:

$$x(\ln 7 - \ln 5) = 3\ln 5 + 2\ln 7 \implies x = \frac{\ln 5^3 + \ln 7^2}{\ln(7/5)} = \frac{\ln 6125}{\ln(7/5)}.$$

TIP *Give an exact answer, like* $x = \dfrac{\ln 6125}{\ln(7/5)}$*, whenever a question says exact or on a non-calculator paper (Paper 1). When a decimal is asked for, or on a calculator paper (Paper 2), round to three significant figures unless told otherwise. Three significant figures is the IB default, and rounding too early or giving too few figures costs accuracy marks.*

FROM words to an equation

Many growth and decay problems use one model,

$$A(t) = A_0 \times r^t,$$

where A_0 is the starting amount and r is the multiplier for each period. The wording gives you r. A rise of 6% per year means $r = 1.06$. A fall of 6% means $r = 0.94$. A quantity that doubles each period has $r = 2$. When the change is continuous rather than in steps, the model uses base e:

$$A(t) = A_0\, e^{kt}.$$

Once the model is written, the logarithm provides the final step.

EXAMPLE 1.16

Sea ice in a Finnish bay covers 1200 km^2 and shrinks by 6% each year. After how many years will it have shrunk to 800 km^2?

Each year multiplies the area by 0.94, so after t years the area is $A(t) = 1200(0.94)^t$. Set this equal to the target and solve:

$$1200(0.94)^t = 800 \implies (0.94)^t = \tfrac{2}{3} \implies t = \frac{\ln(2/3)}{\ln 0.94} \approx 6.55.$$

The ice passes 800 km^2 partway through the seventh year.

EXAMPLE 1.17

Algae in a Finnish lake grow continuously at 12% per year, starting from 200 grams. When does the mass reach 500 grams?

Continuous growth uses base e, so $A(t) = 200\, e^{0.12t}$. Set it equal to the target:

$$200\, e^{0.12t} = 500 \implies e^{0.12t} = 2.5 \implies t = \frac{\ln 2.5}{0.12} \approx 7.6 \text{ years}.$$

The 12% here is a continuous rate, so it sits in the exponent as 0.12 directly, with no conversion to a yearly factor.

EXAMPLE 1.18

A medical isotope has a half-life of 6 hours: every 6 hours, half of it decays. An 80 mg sample is prepared. How long until only 12 mg remains?

Halving every 6 hours means $A(t) = 80\left(\frac{1}{2}\right)^{t/6}$, with t in hours. Set this equal to the target and take a logarithm:

$$80\left(\tfrac{1}{2}\right)^{t/6} = 12 \implies \left(\tfrac{1}{2}\right)^{t/6} = 0.15 \implies t = 6 \cdot \frac{\ln 0.15}{\ln 0.5} \approx 16.4 \text{ hours.}$$

Every half-life problem has this form: the base is $\frac{1}{2}$, and the exponent is the elapsed time divided by the half-life.

YOUR TURN 1.3 The Logarithm: an Exponent Reversed

15. Evaluate each of the following without a calculator.

(a) $\log_2 32$ [1]

(b) $\log_3 \dfrac{1}{9}$ [2]

(c) $e^{2\ln 3}$ [2]

(d) $\ln e^5$ [1]

16. Rewrite each statement in the other form.

(a) Write $3^4 = 81$ in logarithmic form. [1]

(b) Write $\log_5 x = 3$ in exponential form, and hence find x. [2]

17. Solve each equation.

(a) $\log_4(x+1) = 2$ [2]

(b) $\log_2 x + \log_2(x-2) = 3$ [4]

18. Write each expression as a single logarithm.

(a) $\log_3 5 + \log_3 7$ [1]

(b) $\log_2 24 - \log_2 3$, and evaluate it [2]

(c) $3\log_a x + \log_a y$ [2]

(d) $2\ln a - 3\ln b + \ln c$ [2]

19. Express each logarithm in terms of the quantities given.

(a) $\log_5\left(\dfrac{a^2}{b}\right)$, in terms of $\log_5 a$ and $\log_5 b$ [2]

(b) $\log_3 18$, given $p = \log_3 2$ [3]

(c) $\log_{10} 72$, given $\log_{10} 2 = p$ and $\log_{10} 3 = q$ [3]

(d) $\log_5 75$, given $\log_5 3 = 0.682$ [3]

(e) $\log_4 9$, in terms of natural logarithms [1]

(f) $\log_8 x$, in terms of $\log_2 x$ [2]

20. Show that each statement is true.

(a) $\log_a(x^2-1)-\log_a(x+1)=\log_a(x-1)$, for $x>1$ [2]

(b) $\log_a b=\dfrac{1}{\log_b a}$ [2]

21. GDC Evaluate $\log_3 50$ to 3 significant figures. [2]

22. Solve each equation, giving your answer as an exact value.

(a) $5^{2x}=125$ [2]

(b) $4^x=8^{x-1}$ [3]

23. GDC Solve each equation, giving your answer to 3 significant figures.

(a) $2^x=10$ [2]

(b) $3^{x+1}=20$ [2]

24. Solve each equation. *(Hint: in (a) and (b), substitute u for the base raised to x)*

(a) $e^{2x}-5e^x+6=0$ [4]

(b) $2^{2x}-6\cdot 2^x+8=0$ [4]

(c) $3\cdot 2^x=2\cdot 3^x$ [3]

Chapter 1 · Exponents & Logarithms

Reflections

TOK John Napier spent twenty years computing logarithms by hand and published his tables in 1614, not out of mathematical curiosity, but to help sailors and astronomers multiply enormous numbers. He never used the word "logarithm" to mean what we mean today. Does mathematics develop because we need it, or do we find uses for what mathematicians have already built?

TOK The exponent laws force $a^0 = 1$ for every $a \neq 0$, but 0^0 is left undefined, because approaching it along different routes gives different answers. Does leaving 0^0 undefined reveal a limitation of mathematics or the precision of mathematics?

IM Base-10 logarithms feel natural to us because we count in base 10. But the choice is cultural, not mathematical. The Babylonians used base 60 (traces of which survive in our 60-minute hour and 360-degree circle). What does it mean for a mathematical tool to be "natural"?

IM The word *algebra* comes from *al-jabr*, a technique for solving equations described by the Persian mathematician al-Khwarizmi in ninth-century Baghdad; the word *algorithm* is a Latinisation of his own name. His treatises on equations, and the work of mathematicians across the wider Islamic world who followed him, were translated into Latin centuries later and became a foundation of European algebra, including the exponent notation used throughout this chapter. The rules you are applying here developed over many centuries and across many cultures.

RW The decibel (dB) scale for sound is logarithmic: the sound level is $10\log_{10}(I/I_0)$, comparing a sound's intensity I to a fixed reference intensity I_0, so each increase of 10 dB means ten times the intensity. The stellar magnitude scale in astronomy and the octave system in music work the same way. In each case, human perception responds approximately to the logarithm of the physical quantity, not to the quantity itself.

EXT Euler's identity: $e^{i\pi} + 1 = 0$. Five fundamental constants, one equation. It is the exponential function extended into the complex plane, connecting the circular motion of $e^{i\theta} = \cos\theta + i\sin\theta$ to the real exponential growth you have been studying here.

End-of-Chapter Problems — Chapter 1: Exponents & Logarithms

1. Solve $3^{2x+1} = 27^{x-1}$. [3]

2. Show that the equation $2^{x+3} - 2^x = k$ has a real solution only when $k > 0$, and find this solution in terms of k. [5]

3.

(a) Simplify $(2x^{-2}y^3)^3 \times (4x^4y^{-1})^{-1}$, giving your answer with positive indices. [3]
(b) Solve $8^x = 2^{x+6}$. [2]
(c) Show that $\left(a^{1/2} + a^{-1/2}\right)^2 = a + 2 + a^{-1}$, for $a > 0$. [2]

4. GDC The mass of the Earth is 5.97×10^{24} kg and the mass of the Moon is 7.35×10^{22} kg.

(a) Find how many times heavier the Earth is than the Moon, to 3 significant figures. [2]
(b) Find their combined mass, giving your answer in standard form to 3 significant figures. [2]

5. Prove that $\log_a b \cdot \log_b a = 1$ for $a, b > 0$, $a, b \neq 1$. Hence simplify $(\log_2 3)(\log_3 8)$. [4]

6. Given that $\log_a x = 3$ and $\log_a y = 5$, find each of the following.

(a) $\log_a (x^2 y)$ [1] (b) $\log_a \sqrt{xy}$ [2]
(c) $\log_a \left(\dfrac{x}{y^2}\right)$ [1]

7. Let $\log_a 2 = p$ and $\log_a 3 = q$.

(a) Write $\log_a 6$ in terms of p and q. [1]
(b) Write $\log_a \sqrt{12}$ in terms of p and q. [2]
(c) Write $\log_a \dfrac{4}{9}$ in terms of p and q. [2]

8.

(a) Show that $\log_a \sqrt[n]{x} = \dfrac{1}{n} \log_a x$. [2]
(b) Hence solve $\log_2 \sqrt{x} + \log_4 x = 6$. [4]

9. Prove that $\log_a b \cdot \log_b c \cdot \log_c a = 1$ for all valid bases and arguments. Hence evaluate $\log_2 3 \cdot \log_3 5 \cdot \log_5 8$ without a calculator. [6]

10. Given that $\log_2 3 = a$, express each of the following in terms of a:

(a) $\log_2 9$ [1]
(b) $\log_4 3$ [2]
(c) $\log_8 27$ [2]
(d) $\log_3 8$ [2]

11. Solve $\log_3(x+5) - \log_3(x-1) = 2$. [4]

12. Solve $\log_6(x+2) + \log_6(x-3) = 1$, justifying any rejection of solutions. [4]

13. Solve the simultaneous equations:

$$\log_2 x + \log_2 y = 5, \qquad \log_2(x-y) = 3.$$

[5]

14. Solve $\log_x 8 = \log_4 x$. [5]

15. GDC Solve $\log_2(x^2 - 2x) = \log_4(x+4)$. [5]

16. GDC

(a) Express $\log_3 x - 2 = \log_9(x+6)$ as a quadratic equation in x, using change of base. [3]
(b) Hence find the value of x. [2]

17. Solve $\log_5(x^2 - 2x + 1) = 2\log_5(x-1) + 1$, or show that no solution exists. [5]

18. Solve $7^{x-2} = 5^{x+1}$, giving your answer in exact logarithmic form. [4]

19. Find the exact solution to $e^{2x} = 3e^x + 10$. *(Hint: let $u = e^x$)* [4]

20. Solve the equation $9^x - 4 \cdot 3^{x+1} + 27 = 0$. *(Hint: let $u = 3^x$; note $9^x = u^2$)* [5]

21. The number of bacteria in a culture doubles every 3 hours. Initially there are 500 bacteria.

(a) Write an expression for the number of bacteria after t hours. [2]
(b) Find the time, to the nearest minute, when there are 50 000 bacteria. [3]

22. GDC The population of a city is modelled by $P(t) = 250\,000 \times e^{0.032t}$, where t is the number of years after 2020.

(a) Write down the population in 2020. [1]
(b) Find the population in 2030, to the nearest thousand. [2]
(c) Find the year in which the population first exceeds 400 000. [3]

23. GDC The value V of a car after t years is given by $V = 32\,000 \times (0.85)^t$.

(a) Write down the initial value of the car. [1]
(b) Find the value after 5 years, to the nearest dollar. [2]
(c) Find the number of years until the value falls below \$10 000. [3]

24. GDC The pH of a solution is defined by pH $= -\log_{10}[\mathrm{H}^+]$, where $[\mathrm{H}^+]$ is the hydrogen ion concentration in mol/L.

(a) A solution has $[\mathrm{H}^+] = 3.5 \times 10^{-4}$ mol/L. Find its pH to 2 decimal places. [2]
(b) A solution has pH $= 8.3$. Find its hydrogen ion concentration in standard form. [2]
(c) Solution A has pH $= 3$ and solution B has pH $= 7$. Find how many times more acidic solution A is than solution B. [2]

25. Let $f(x) = \log_3(2x-1)$ and $g(x) = 3^x + 2$. (Inverse functions and graph transformations are developed in Ch. 6; this question looks ahead to them.)

(a) Find the domain of f. [1]
(b) Find $f^{-1}(x)$. [3]
(c) Show that $f^{-1}(x) = \frac{1}{2}g(x) - \frac{1}{2}$, and describe the transformation of g this represents. [3]

26. GDC A radioactive substance decays so that its mass M grams after t years satisfies $M = M_0 e^{-kt}$, where M_0 and k are positive constants. After 10 years the mass is halved.

(a) Show that $k = \dfrac{\ln 2}{10}$. [2]
(b) Find the percentage of the original mass remaining after 25 years. [3]
(c) Find the time for the mass to reduce to 10% of its original value. [3]

27. GDC This question is about writing very large and very small quantities in standard form.

(a) A human cell has diameter about 2×10^{-5} m, and a water molecule about 2.8×10^{-10} m. Find how many times wider the cell is, giving your answer in standard form to three significant figures. [3]
(b) The observable universe is about 8.8×10^{26} m across, and the shortest length with any physical meaning, the Planck length, is about 1.6×10^{-35} m. Find the ratio of the two, in standard form. [3]
(c) A sheet of paper is 0.1 mm thick. It is folded in half, then in half again, and so on. Find the thickness after 20 folds, in metres and in standard form. [3]
(d) Find the least number of folds needed for the thickness to exceed the distance to the Moon, 3.84×10^8 m. [4]
(e) Your answer to part (d) is far smaller than most people expect. Explain why, referring to how the thickness grows. [2]

28. GDC An amount P is invested. Bank A pays 4% interest compounded annually. Bank B pays 3.9% compounded monthly, so its value after t years is $P\left(1+\frac{0.039}{12}\right)^{12t}$.

(a) Write down an expression for the value in Bank A after t years. [2]

(b) Show that Bank B's value can be written as Pr^t, and find r to five decimal places. [3]

(c) Determine which bank gives more after 10 years, and by how much as a percentage of P. [3]

(d) Use logarithms to find how long an investment in Bank A takes to double. [4]

(e) Bank C pays 4% compounded continuously, giving $Pe^{0.04t}$. Show that Bank C always beats Bank A, and find the difference after 10 years as a percentage of P. [3]

29.

(a) Prove the change of base rule $\log_a b = \dfrac{\log_c b}{\log_c a}$, for valid bases a and c. *(Hint: let $x = \log_a b$, so $a^x = b$, then take $\log_c$ of both sides)* [3]

(b) Hence show that $\log_4 x = \frac{1}{2}\log_2 x$. [2]

(c) Solve $\log_2 x + \log_4 x + \log_{16} x = 7$. [4]

(d) Solve $\log_x 2 + \log_{x^2} 2 = 3$, giving an exact answer. [5]

Challenge Questions

30. *Investigation.* Consider $f(x) = \log_a x$ for different values of $a > 0$, $a \neq 1$.

(a) Show that $\log_{1/a} x = -\log_a x$ for all valid x. [3]

(b) Show that $\log_{a^n} x = \dfrac{1}{n}\log_a x$. [3]

(c) Using part (b), find all values of a for which $\log_a x = \log_{a^2}(x+2)$ has a solution, and find that solution. [5]

31. The loudness L of a sound (in decibels) is given by $L = 10\log_{10}\left(\dfrac{I}{I_0}\right)$, where I is the intensity in $\mathrm{W/m^2}$ and $I_0 = 10^{-12}$ $\mathrm{W/m^2}$ is the threshold of hearing.

(a) A concert has loudness 110 dB. Find the intensity I. [3]

(b) A whisper has loudness 30 dB. Show that the concert is 10^8 times more intense than the whisper. [3]

(c) Two sounds with intensities I_1 and I_2 are played together. Show that the combined loudness is not simply $L_1 + L_2$. Find the combined loudness when two 80 dB sounds are played simultaneously. [4]

32. **Counting digits, and the first digit.** This investigation uses logarithms to count digits, and then to explain a pattern that auditors use to detect faked data.

(a) Let N be a positive integer. Show that N has exactly $\lfloor \log_{10} N \rfloor + 1$ digits, where $\lfloor t \rfloor$ is the greatest integer not exceeding t. [3]

(b) Hence find the number of digits in 2^{1000}. [2]

(c) The first digit of N is d exactly when the fractional part of $\log_{10} N$ lies in $[\log_{10} d,\ \log_{10}(d+1))$. Verify this for $N = 2^7$. [3]

(d) Assume the fractional parts of $\log_{10}(2^n)$ are spread evenly over $[0, 1)$ as n increases. Show that the proportion of powers of 2 whose first digit is d is $\log_{10}\left(1 + \dfrac{1}{d}\right)$. [4]

(e) Evaluate this proportion for $d = 1$ and for $d = 9$. Then count the first digits of $2^1, 2^2, \ldots, 2^{30}$ on your GDC and compare. [5]

(f) The pattern in part (d) is known as Benford's law. Comment on how well your counts in part (e) support it, and state one reason why a sample of 30 might mislead. [3]